Acute COVID-19 Infection in Children

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Introduction

- ☑ Symptomatic infections due to severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2)-related coronavirus disease 2019 (COVID-19) is less frequent in children compared to adults. But infection rate and viral loads in the nasopharynx of both children and adults appear to be similar.
- ☑ The hospitalization rates and severity of illness are significantly lower in children compared to adults. Infants <1 year and children with underlying medical conditions have an increased risk of having a severe illness.

Clinical Features

- ☑ The incubation period can range from 2 to 14 days (average of 6 days).
- ☑ Signs and symptoms include fever, cough, fatigue, headache, myalgia, nasal congestion, loss of taste or smell, sore throat, breathing difficulty, abdominal pain, diarrhea, vomiting, neurological symptoms, etc.
- ☑ Croup, febrile seizures, and exacerbation of wheeze have also been reported following COVID-19 infection in children.

Risk Factors for Severe Disease

- ☑ Children with underlying medical conditions such as neurological impairment, genetic syndromes, and metabolic and congenital heart disease may be at higher risk of severe disease.
- ☑ Most mortality in children has been reported in children with underlying medical conditions and in those <1 year of age.
- ☑ Children with obesity, immunosuppression, chronic lung disease, diabetes mellitus, sickle cell disease, etc., might also be at increased risk of severe disease.

Classification of Severity of Illness

Classification of severity of illness is presented in **Table 1**.

TABLE 1: Classification of severity of illness.

Mild	***Moderate***	***Severe***
Uncomplicated URI	Nonsevere pneumonia (without severe hypoxia or danger signs)	Severe pneumonia/danger signs/ organ dysfunction
Fever, sore throat, rhinorrhea, cough without hypoxia, breathlessness, or fast breathing	SpO_2 90–94% in room air Fast breathing (age based) Respiratory rate <2 months: ≥60 breaths/minute 2–11 months: ≥50 breaths/ minute 1–5 years: ≥40 breaths/minute	SpO_2 < 90% in room air Danger signs like inability to breastfeed or drink, grunting, severe lower chest in drawings, cyanosis, lethargy, somnolence, unconsciousness, seizures, ARDS, MODS, and septic shock

(ARDS: acute respiratory distress syndrome; MODS: multiple organ dysfunction syndrome; SpO_2: saturation of peripheral oxygen; URI: upper respiratory infection)

Classification of Severity of Illness

Mild Disease: Treatment

- ☑ Can be managed at home with symptomatic treatment.
- ☑ Avoid using nonsteroidal anti-inflammatory drugs other than paracetamol.
- ☑ Treat fever with paracetamol 10–15 mg/kg/dose. May be given 4–6th hourly for fever.
- ☑ Ensure adequate feeds and maintain hydration.
- ☑ In case of diarrhea, give oral rehydration solution (ORS) 10 mL/kg/purge.
- ☑ No role for antivirals, antibiotics, or investigations.
- ☑ Maintain COVID appropriate behavior.
- ☑ Counsel patients about danger signs.
- ☑ Occurrence of any new symptom or worsening of existing symptom warrants review.
- ☑ Monitor every 24–48 hours for worsening of symptoms.

Moderate Disease

These are children with features of pneumonia in the form of fast breathing without danger signs. These children should be admitted to hospital.

- ☑ Provide oxygen to target saturation of peripheral oxygen (SpO_2) 94–98%.
- ☑ Steroids should be given only if rapid deterioration and >5 days since onset of disease.
- ☑ Antibiotics may be given if there is a strong suspicion of bacterial infection.
- ☑ *Investigations*: Check blood glucose, complete blood count (CBC), and chest X-ray. *CT chest is not indicated for diagnosis or management of COVID-19 in children.*
- ☑ *Monitoring*: Monitor sensorium, heart rate, respiratory rate, blood pressure, SpO_2, and work of breathing.

Severe Disease

These are children with severe pneumonia with hypoxia, danger signs, or organ dysfunctions. They should be admitted in high dependency unit/pediatric intensive care unit (HDU/PICU).

- ☑ Administer oxygen with an appropriate oxygen delivery device (target saturation 94–98% during resuscitation) **(Table 2)**.

TABLE 2: Oxygen delivery devices and fraction of inspired oxygen (FiO_2) delivered.

Device	***Flow rate***	***FiO_2***
Nasal cannula	1 L/minute	24%
Nasal cannula	4 L/minute	36%
Face mask	>6 L/minute	40–60%
Venturi mask	Based on adapter chosen	24–60%
Nonrebreathing mask	10–15 L/minute	100%
High-flow nasal cannula	1–2 L/kg/minute flow	21–100%

- ☑ Empiric antibiotics should be started, especially if suspicion of secondary bacterial infection present.
- ☑ *Steroids*: Steroids should be used at the right time, in right dose, and for the right duration.
- ☑ Steroids are not indicated and are harmful in asymptomatic and mild cases of COVID-19.
- ☑ Start steroids only 3–5 days after onset of disease as it prolongs viral shedding. It is indicated only in moderate and severe disease with increasing oxygen demand.
- ☑ In children on high-flow nasal cannula (HFNC), high fraction of inspired oxygen (FiO_2), noninvasive ventilation (NIV), invasive ventilation, or extracorporeal membrane oxygenation (ECMO), start steroids at:
  - Dexamethasone 0.15 mg/kg, maximum dose 6 mg once a day or
  - Methylprednisolone 0.75 mg/kg, maximum dose 30 mg once a day
  - Continue for 5–7 days and taper, up to 14 days, depending on clinical assessment on daily basis.
- ☑ *Remdesivir*: Guidelines from Ministry of Health and Family Welfare, Government of India does not recommend use of remdesivir in children with COVID-19 infection.
- ☑ Newer evidence states that it may be beneficial when given early in the course of disease in patients with risk factors for severe disease and have an emergent or increasing need for supplemental oxygen.
- ☑ A 3-day course of remdesivir early in the disease was found to cause 87% relative reduction in admission and death. There is no role for remdesivir beyond 10 days of infection. It can cause hepatic derangement, hence avoid in children with hepatic derangement.

Severe Disease

- ☑ The Centers for Disease Control and Prevention (CDC) recommendations in children are based on outcome and safety data for adults and child's risk of disease progression. The CDC recommended indications include:
  - Hospitalized children aged ≥12 years with COVID-19 who have risk factors for severe disease and have an emergent or increasing need for supplemental oxygen (moderate evidence).
  - Hospitalized children aged ≥16 years with COVID-19 who have an emergent or increasing need for supplemental oxygen regardless of whether they have risks factors for severe disease (moderate evidence).
  - In consultation with pediatric infectious disease specialist, remdesivir may be considered for children of all ages with COVID-19 who have emergent or increasing need for supplemental oxygen (weak evidence).

  Dose: 5 mg/kg IV (maximum 200 mg) loading dose over 30–120 minutes on day 1 followed by 2.5 mg/kg (maximum 100 mg) IV OD on days 2–4.
- ☑ *Shock*: If patient is in shock, resuscitate with isotonic crystalloid boluses (normal saline/Ringer's lactate/Plasmalyte) at 10–20 mL/kg over 30–60 minutes. Titrate inotropes if unresponsive to fluids. Posthemodynamic stabilization follows a conservative fluid strategy to avoid fluid overload.
- ☑ Administer antibiotics according to hospital antibiogram within first hour of admission after taking blood cultures.
- ☑ *Transfusion trigger:* Hemoglobin < 7 g/dL for children with stable oxygenation and hemodynamics and <10 g/dL for those with refractory hypoxemia or shock. For fluid refractory catecholamine-resistant shock hydrocortisone 2–4 mg/kg/24 hours may be added. (Avoid if already on dexamethasone or methylprednisolone).
- ☑ *Acute respiratory distress syndrome (ARDS)*: In case of mild ARDS, HFNC or NIV may be tried. If patient is having moderate-to-severe ARDS, provide invasive ventilation. Use low tidal volume (4–8 mL/kg) ventilation with optimum positive end-expiratory pressure (PEEP).
- ☑ Aim to keep P plateau <28 cmH_2O and driving pressure <15 cmH_2O. Use of cuffed endotracheal (ET) tube, in line suction with minimal disconnection of ventilator circuits and early proning should be considered in children with severe ARDS.
- ☑ Provide appropriate sedation and analgesia and restrict fluids with the aim to keep fluid overload at <10%.
- ☑ *Transfusion trigger:* Hemoglobin < 7 g/dL for children with stable oxygenation and hemodynamics and <10 g/dL for those with refractory hypoxemia or shock.
- ☑ Provide organ support like renal replacement therapy when indicated.
- ☑ *Investigations*: CBC, blood glucose, renal function test (RFT), liver function test (LFT), C-reactive protein (CRP), ferritin, D-dimer, and chest X-ray.

Severe Disease

- ☑ *Prophylactic anticoagulation*: Not routinely recommended in children. All hospitalized children should be monitored for thrombosis **(Table 3)**. The decision to administer prophylactic anticoagulation must be balanced with the child's bleeding risk. On suspicion of thrombosis, confirm by appropriate investigations and provide anticoagulation with low molecular weight heparin for 12 weeks.

TABLE 3: Risk factors for thrombosis.

☑ Personal history of venous thrombotic events (VTE)	☑ Burns
☑ Family history of first-degree relative with VTE	☑ Active malignancy
☑ Presence of central venous line	☑ Estrogen therapy
☑ Decreased mobility from baseline	☑ Flare of inflammatory disease
	☑ Morbid obesity
	☑ Severe dehydration
	☑ Recent surgery or trauma

Indications for prophylactic anticoagulation in admitted patients:

- Strong personal or family history of venous thromboembolism (VTE), or
- An indwelling central venous line and two or more additional risk factors, or
- Four or more risk factors
- D-dimer more than five times the upper limit of normal in an admitted child
- Children already on anticoagulation therapy may continue same unless they develop active bleeding.

- ☑ *Low molecular weight heparin*: Use in children without severe renal impairment with risk factors for thrombosis and severe COVID-19 infection.
  - Prophylactic dose 0.5 mg/kg twice daily, till child is discharged from hospital (anti-Xa level 0.2 to <0.5 U/mL)
  - Therapeutic dose in case of documented thrombosis: 1 mg/kg twice daily.

Further Reading

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Developmental Dysplasia of the Hip

Lead Author

Ashima Sahetiya

Co-Authors

SN Prashanth, Devender Gaba

Under the Auspices of the IAP Action Plan 2022

Remesh Kumar R
IAP President 2022

Upendra Kinjawadekar
IAP President-Elect 2022

Piyush Gupta
IAP President 2021

Vineet Saxena
IAP HSG 2022–2023

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Chairperson

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